

5. Project Development Phase

Source Code

The complete source code is available in the root directory:

- `/backend` — FastAPI backend application
- `/frontend` — React + Vite frontend application

Key Modules

Backend

- `main.py` — API routes and application entry point
- `models.py` — SQLAlchemy database models (User, DebtRecord)
- `schemas.py` — Pydantic request/response schemas
- `auth.py` — Password hashing and JWT token generation
- `ai_engine.py` — Google Gemini AI integration functions
- `database.py` — Database connection and session setup

Frontend

- `App.jsx` — Landing/auth flow routing
- `Landing.jsx` — Marketing landing page
- `Dashboard.jsx` — Main application (dashboard, loans, history, profile)

API Endpoints Implemented

Method	Endpoint	Purpose
POST	<code>/register</code>	Create new user account
POST	<code>/login</code>	Authenticate and receive JWT
POST	<code>/debt-record</code>	Add new loan with AI strategy
GET	<code>/loans</code>	List user’s loans
DELETE	<code>/loans/{id}</code>	Delete a loan record
GET	<code>/financial-health</code>	Get financial health summary
POST	<code>/negotiate</code>	Get AI negotiation strategy
POST	<code>/settlement-predictor</code>	Get AI settlement prediction
POST	<code>/generate-negotiation-email</code>	Get AI-drafted email
GET	<code>/dashboard-data</code>	Aggregated dashboard data